Book Market Analysis Report

Project Overview

This project involved web scraping book data from books.toscrape.com, cleaning and transforming the data, performing a market analysis, and visualizing key insights.

Data Acquisition & Preparation

We successfully scraped information for 20 books from the first page of the target website. The collected data included book titles, prices, and stock status. During preparation, prices were converted to Indian Rupees (INR) for local relevance, and stock status was simplified into a boolean 'Is_Available' column.

Key Findings

1. Price Analysis by Rating

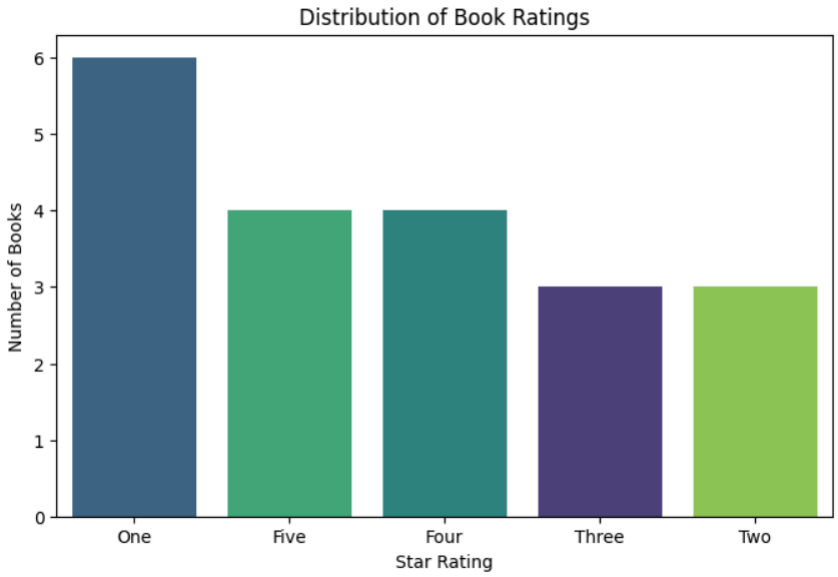
Our analysis revealed interesting patterns in pricing across different book ratings:

Rating	Average Price (£)	Count
Five	39.75	4
Four	31.11	4
One	40.02	6
Three	42.32	3
Two	36.83	3

- Books rated 'Three' have the highest average price, while 'Four'-star books have the lowest average price on this page.
- 'One'-star rated books are the most numerous in this sample.

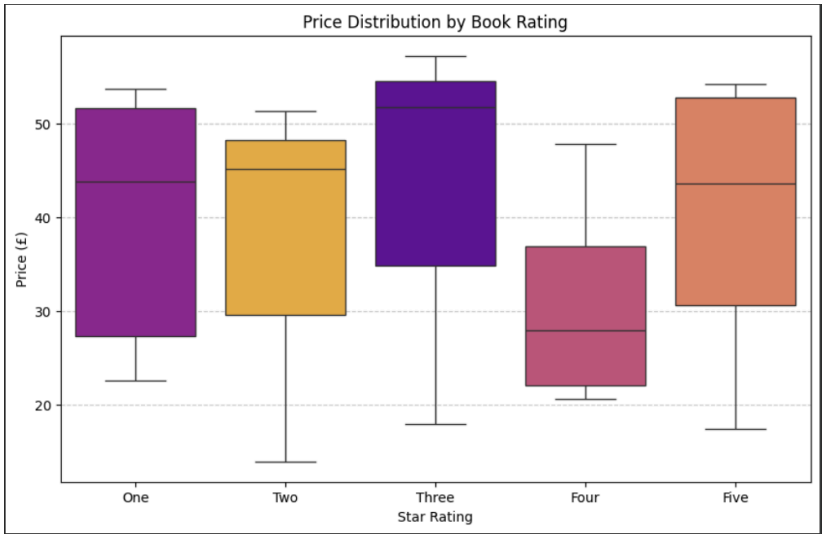
2. Visualizations

- **Distribution of Book Ratings**: The count plot shows the frequency of each star rating, highlighting which ratings are more common among the scraped books.

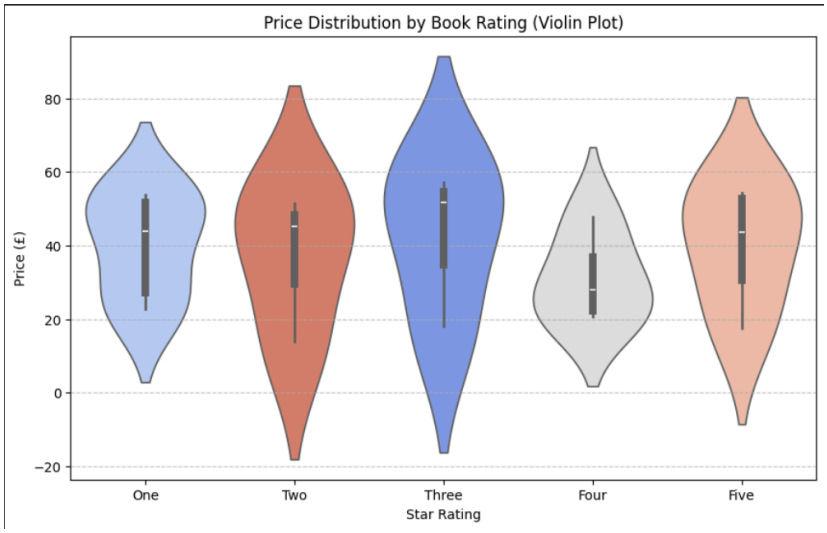


Countplot

- **Price Distribution by Book Rating (Box Plot & Violin Plot)**: These plots illustrate the spread of prices within each rating category. They help identify median prices, interquartile ranges, and potential outliers, offering a detailed view beyond just the average.

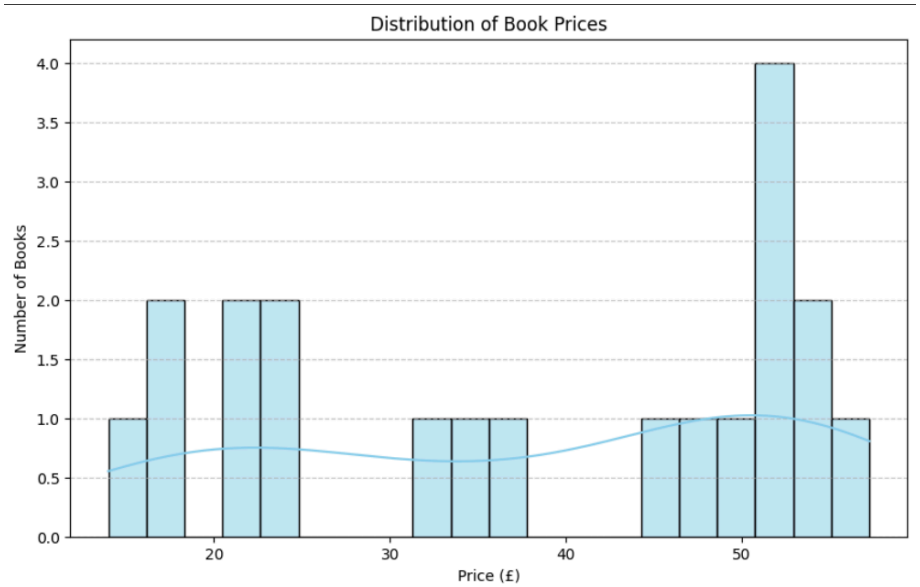


Box Plot



Violin Plot

- **Distribution of Book Prices (Histogram)**: The histogram provides a clear view of the overall price frequency, showing how many books fall into various price ranges and indicating the most common price points.



Histogram

Conclusion

This analysis provides an initial snapshot of book pricing and rating distributions on books.toscrape.com. The insights gathered can be further explored to understand market dynamics, pricing strategies, and customer preferences.

Data Export

All raw scraped data and the aggregated price analysis have been exported to an Excel file named Market_Analysis_Report.xlsx for further detailed review.